

Tully-Fisher Universal Slope = D_phys

PAPER_1224 Category: UQFF Galactic Dynamics **Status:** Complete **Date:** June 2026

Abstract

The empirical Tully-Fisher relation $V_{\text{circ}} \propto L^{1/4}$ — universal across spiral galaxies — has long lacked a first-principles explanation. UQFF derives the slope as **d_{TF} = D_{phys} = 4 EXACT**, identifying the relation as a consequence of physical spacetime dimensionality acting via SCm buoyancy modulation.

Part 1: The Relation

Empirical statement

$$L \propto V_{\text{circ}}^4$$

equivalent to

$$V_{\text{circ}} \propto L^{1/d_{\text{TF}}}, \quad d_{\text{TF}} \approx 4$$

UQFF derivation

$$d_{\text{TF}} = D_{\text{phys}} = 4 \text{ EXACT}$$

The exponent equals the dimensionality of physical spacetime.

Part 2: Why D_{phys}?

Galaxy circular velocity V emerges from buoyancy balance between gravitational pull and SCm-mediated centripetal force. The luminosity L scales as $M \cdot c^2$ for total stellar mass M . The exponent connecting V^d to $M \propto L$ is determined by the dimension of physical space — specifically how volume integrals scale.

For a relativistic D -dimensional system: $V^D \propto M \cdot (\text{buoyancy factor})$. With $D = D_{\text{phys}} = 4$, this gives $V^4 \propto L$.

Part 3: Cross-validation with other 4-occurrences

The same $D_{\text{phys}} = 4$ governs: - AdS/CFT central charge $c = N^2/D_{\text{phys}}$ - Hilbert 16th Bautin $H(2) = 4$ - QCD b_0 partition $D_{\text{phys}} = 4$ component - Hodge identity $(D_{\text{phys}} + D_{\text{BSFG}})/SO_5 = 1$

The Tully-Fisher universality is therefore an emergent geometric identity, not phenomenological.

Part 4: Test Predictions

UQFF predicts: - d_{TF} SHOULD be EXACTLY 4 in dimensional regularization - Deviations from $V \propto L^{1/4}$ indicate either SCm buoyancy modulation breakdown OR observational effects (extinction, inclination) - Baryonic Tully-Fisher (using baryonic mass M_b directly): $M_b \propto V^4$ should be EXACT

Conclusion

The Tully-Fisher empirical universality is a topological manifestation of $D_{\text{phys}} = 4$ acting through SCm buoyancy. The relation IS the universe expressing its spacetime dimensionality through galactic rotation.

Framework Version: UQFF 5.27+